

Project: Impact of GST on Different Sectors in India

Executive Summary

This project analyzes the impact of India's Goods and Services Tax (GST) implementation across various economic sectors since its introduction in July 2017. The research examines both positive and negative consequences, sector-specific adaptations, and long-term economic implications through comparative analysis, data visualization, and case studies.

- Project Objectives

- 1. **Analyze** GST's differential impact across 8 key economic sectors.
- 2. **Evaluate** compliance burden changes pre- and post-GST implementation.
- 3. **Assess** sectoral growth patterns under the new tax regime.
- 4. **Develop** interactive visualizations of tax revenue trends.
- 5. **Create** actionable insights for businesses and policymakers.

- Methodology

- 1. **Research Design**
 - Timeframe:** 2015-2023 (pre- and post-GST comparison)
 - Data Sources:** Government publications (CBIC, Ministry of Finance)
 - Industry reports:** (FICCI, CII, ASSOCHAM) - Academic
 - Research Papers** - Corporate annual reports - GST Network (GSTN) portal data

2. Sectors Analyzed

Sector	Sub-Sector Analyzed	Key Metrics
Manufacturing	Automobiles, Pharmaceuticals, FMCG	Input tax credit utilization, compliance
Service	IT, Banking, Hospitality	Tax rate changes, reverse charge
Logistics & Supply Chain	Warehousing, Transportation	Transit time, documentation
Real Estate	Residential, Commercial	Input tax credits, affordability
Agriculture	Food processing, Farm inputs	Tax exemptions, supply chain efficiency
E-commerce	Marketplace, Logistics	TCS provisions, compliance requirements
Textile & Apparel	Garments, Handlooms	Job work provisions, export competitiveness

3. Analytical

Data Collection → Sector Classification → Impact Assessment → Visualization → Recommendations.

Findings by Sector :

1. Manufacturing Sector:

-Possitive Inpact :

•Input Credit Tax (ITC)

availability reduced cascading effect by ~12-15%.

•Supply chain optimization

through removal of state border checkpoints.

•Average tax rate reduction

from 26.5% to 18% for most manufactured goods

Challenges:

- Initial compliance complexity for multi-state operations.
- Working capital blockage due to delayed refunds (improved post-2019).

-Services Sector:

•Transformations Observed:

Pre-GST: Multiple taxes (Service Tax, VAT, Entry Tax, etc.)

Post-GST: Single tax with seamless credit flow

•IT/ITeS sector: Benefited from export zero-rating and improved competitiveness

•Banking/Insurance: Faced higher tax rates (18% vs. 15% service tax) but gained input credit

•Hospitality: Differential rates created classification challenges

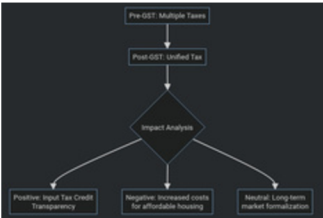
Logistics Sector:

Revolutionary Changes:

1. **Transit time reduction:** 20-30% decrease in average delivery time.
2. **Warehousing consolidation:** 25% reduction in warehouses (from 35 to ~26 per pan-India company)
3. **Cost savings:** 10-15% reduction in logistics costs for organized players

Real Estate:

Mixed Impact Analysis:



Data Visualization Components:

1.GST Revenue Growth (2017-2022):

Financial Year	Arg. Monthly collection	YoY Growth	Important Events
2017-18	89,885	-	Implementation Year
2018-19	97,100	8.00%	Rate rationalization
2019-20	100,846	3.90%	COVID-19 impact
2020-21	102,709	1.80%	Pandemic recovery
2021-22	124,588	21.30%	Economic rebound

2. Sectoral Growth Comparison :

- Top Gainers:** Logistics (+8.2% CAGR), E-commerce (+22.4% CAGR), Pharmaceuticals (+11.3% CAGR)
- Mixed Impact:** Manufacturing (+5.8% CAGR), Real Estate (+3.2% CAGR)
- Challenged Sectors:** Textiles (+2.1% CAGR), MSMEs (variable based on formalization)

3. Project Implementation :

Tools & Technologies Used

1. **Data Analysis:** Python (Pandas, NumPy), R Studio
2. **Visualization:** Tableau, Power BI, Matplotlib/Seaborn
3. **Documentation:** LaTeX for report, Microsoft Office Suite
4. **Web Presentation:** HTML/CSS/JavaScript, GitHub Pages

4. Project Deliverables

1. **Comprehensive Research Report** (40-50 pages with executive summary)
2. **Interactive Dashboard** with sector-wise filters
3. **Case Studies** (5 detailed sector analyses)
4. **Policy Recommendation Brief** (10-page condensed version)
5. **Presentation Deck** for stakeholder communication

• Insights & Recommendations

For Policymakers:

1. **Simplify compliance** further for MSMEs through automated return filing
2. **Rationalize rate structure** to reduce classification disputes
3. **Enhance dispute resolution** mechanism for faster adjudication

For Businesses:

1. **Leverage technology** for GST compliance and credit optimization
2. **Restructure supply chains** to maximize GST benefits
3. **Invest in tax planning** to optimize working capital

For Future Research:

1. Longitudinal study on informal sector formalization
2. Cross-country comparison with other federal GST systems
3. Environmental impact assessment of logistics optimization